Linear A

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Summary

Linear A is a Bronze Age (*c.* 1800–1450 BCE) script attested primarily on Crete but also sporadically in the Aegean islands, mainland Greece, and Asia Minor. Typologically it is classified as a logo-syllabary since it consists of signs representing both syllables (syllabograms) and real-world referents (logograms/ideograms). To date, the script, which was used to write the still poorly understood Minoan language, remains undeciphered. Linear A seems to have been used for both administrative and cultic purposes: incised clay documents (tablets, roundels, and sealings) were used in palace administration to record economic transactions, while inscribed carved-stone and metal objects and painted clay vessels have been found in non-administrative contexts, mostly cultic or utilitarian. There is no evidence of Linear A's use in monumental inscriptions, diplomatic correspondence, historiography, or other forms of literature. Still, Linear A is likely to have been used for writing on perishable material (papyrus or parchment) as well, although no examples have survived. Although the script remains undeciphered, some information—place names, personal names, names for commodities, and terms for various sorts of transaction—can still be gleaned from the available texts. Nevertheless, the nature of our evidence (short formulaic inscriptions with limited syntax), the relatively small number of inscriptions that have survived, and their often poor state of preservation significantly hamper our understanding of the language.

Keywords: Linear A, Linear B, Aegean scripts, Bronze Age, Crete, Minoan

Subjects: Greek Material Culture: Bronze Age, Linguistics

Linear A is a logo-syllabic script used in the Bronze Age Aegean *c.* 1800–1450 BCE. The script, still undeciphered, was used to write the Minoan language associated with the Minoan civilization of Bronze Age Crete. Attested primarily on Crete, the script is also found sporadically in the Aegean islands (on Thera, Kea, Melos, Samothrace, and Cythera), mainland Greece (at Haghios Stephanos), and Asia Minor (at Miletus). The extant evidence amounts to *c.* 1,500 inscriptions (some of which are still unpublished) occurring in a variety of administrative and non-administrative contexts. Small clay tablets, roundels, and sealings of various types were used as economic records by palace administrations, while other stone or metal objects bear inscriptions with a ceremonial or dedicatory function. A few graffiti have also survived, although for the most part they are badly preserved and illegible. The script received the name Linear A from Sir Arthur Evans, the excavator of the palace of Knossos, to distinguish it (together with Linear B) from the more pictographic script he dubbed Cretan Hieroglyphic. Like Cretan Hieroglyphic (which is likewise undeciphered), the Linear A script was probably an indigenous creation developed on Crete independently of pre-existing templates but inspired (according to the so-called stimulus diffusion theory) by awareness of the existence of writing in the neighbouring regions of Egypt

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and the Near East. Although we know that Linear A served as a template for the creation of Linear B (which was used to write Mycenaean Greek) and probably for the undeciphered Cypro-Minoan script as well, the precise relationship between Linear A and Cretan Hieroglyphic is still debated.[1]

The Linear A Sign Repertory

Structural Characteristics

Typologically, Linear A is a logo-syllabary, that is, it consists of phonetic signs standing for syllables (syllabograms) and more or less realistic picture-signs standing for commodities (logograms/ideograms). Syllabograms are understood to represent open syllables of the types vowel-only (e.g., /a/), consonant-vowel (e.g., /na/), or consonant-semiconsonant-vowel (e.g., /nwa/). The traditional sign list of Linear A classifies signs based on their graphic form in increasing order of complexity, dividing them into 178 "simple" signs (grid 1 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=22>, grid 2 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=23>) and 164 "complex/composite" signs (grid 3 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=24>, grid 4 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=25>, grid 5 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=26>). In addition, there are 47 fractional signs (grid 6 <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=5&ce=na45u6v17d4od89umoq8mt720858q5r7&sp=27>) and various signs used to write numerals (from units up to thousands).[2] Simple signs are independent graphic units, whereas complex/composite signs are a combination of two or more simple signs by ligature or juxtaposition. Functionally, simple signs are generally used to represent syllables within words (e.g., *ku-ro*, "total") but can behave both syllabically and logographically according to context. By contrast, complex/composite signs only behave logographically/ideographically, although a few may have been read as words (e.g., A 559, which is a combination of AB 80/ma and AB 26/ru, is likely to have been read as /maru/, "wool").[3] When behaving logographically/ideographically, both simple and complex/composite signs stand for groups of people or commodities (e.g., agricultural products, livestock, vessels, textiles). A core group of about ninety simple signs is common to all sites that have yielded Linear A evidence, and these signs are also shared, for the most part, with Linear B. In the sign list, signs common to both Linear A and Linear B are indicated by the prefix "AB." The remaining simple signs, specific to Linear A, are preceded by the prefix "A" and have been found only at a limited number of sites.[4] Complex/composite signs are also specific to Linear A and in fact are mostly "hapaxes" (i.e., they occur only once in the surviving Linear A texts), but the means by which

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simple signs are combined to form complex/composite signs are common across sites.[5] In addition to their syllabic and logographic/ideographic functions, Linear A signs can also function as "transaction signs," providing information about the nature or purpose of a transaction.[6] Transaction signs usually occur in a tablet's heading, enclosed within dots and followed by logograms. Linear A inscriptions are written left to right (although there are a few examples of right-to-left and boustrophedon writing), and words or sign-groups may be separated by a dot. On tablets, words can be split across consecutive lines, and lines are usually not ruled to guide writing.

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Figure 1. Different types of Linear A signs on tablet HT 9a <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=1&ce=540ecor2uudeumpibfksblm7ss9s2lsi&sp=54>: syllabograms (blue), logogram (green), transaction sign (red), numbers (black), fractions (orange), and word-dividers/dots (purple).

Source. Drawing by author.

Relationships between Scripts and the Readability of Linear A

Linear A (which belongs to an unidentified language family) was used as the graphic template for the development of Linear B (which was used to write Greek, an Indo-European language).[7] Graphically, the two scripts show a 70 percent similarity with respect to syllabic signs, but Linear A complex/composite signs do not exist in Linear B and the logographic repertory shows substantial changes. Linear B has been successfully deciphered,[8] and, given the close relationship between the scripts, it is assumed that most homomorphic signs (i.e., signs that look the same in both Linear A and Linear B) may be read with identical, or very similar, phonetic values in both scripts <http://www.people.ku.edu/~jyounger/LinearA/ABgrids.html>.[9] This assumption is supported by evidence of shared sign-sequences in both scripts that seem especially likely to contain homophonic signs based on the meanings of those sign-sequences in Linear B (e.g., place names: *pa-i-to* ["Phaistos"], *su-ki-ri-ta*; and personal names: *da-i-pi-ta*, *pa-ra-ne*). That Linear B was used exclusively for administrative purposes and was likely created on Crete also argues for both graphic and phonetic continuity between the two scripts. It is therefore possible to "read" Linear A inscriptions phonetically (and complete transcriptions have been produced for inscriptions from Haghia Triada <http://people.ku.edu/~jyounger/LinearA/HTtexts.html> and elsewhere <http://people.ku.edu/~jyounger/LinearA/misctexts.html>), although an understanding of what the words mean still eludes us.

Chronological and Geographical Distribution

Most Linear A evidence is dated to the end of the Neo-Palatial period (*c.* 1600–1450 BCE, Late Minoan IA–B),[10] when occasional fires significantly damaged Minoan palaces and baked the clay documents stored in them. In the Neo-Palatial period, Linear A is well attested throughout Crete for both administrative and non-administrative purposes. Archival documents come mostly from Haghia Triada, Khania, Zakros, Phaistos, Knossos, Malia, Arkhanes, and Tylissos. Outside Crete, Linear A is attested in the Aegean islands (Thera, Kea, Melos, Samothrace, and Cythera) in both administrative and non-administrative contexts. It is also attested in Asia Minor (appearing on clay vessels from Miletus) and in mainland Greece (in non-administrative inscriptions from Haghios Stephanos, Tiryns, and Argos). The earliest attestations of Linear A go back to the Proto-Palatial period (*c.* 1800–1700 BCE, Middle Minoan II–III) when Linear A was used concomitantly with Cretan Hieroglyphic.[11] Although the earliest Linear A seems to have been written in south-central Crete (at Phaistos), with Cretan Hieroglyphic being used in northern Crete (at Knossos, Malia, and Petras), there are a few documents from Knossos and Malia (traditionally called "dubitanda <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=31&issue_number=0&ce=cooqm3a5q3s12

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vmvdqppfim4sk80ag09&sp=24>”) that defy clear-cut classification as they could be taken to be either Cretan Hieroglyphic or Linear A.[12] Linear A ceased to be used after the end of the Neo-Palatial period, at least for administrative purposes. A number of objects with Linear A inscriptions have been found in slightly later contexts: these include a figurine from Poros, a building block in the Kephala tholos tomb, a miniature cup from Khania, and a pithoid jar from Knossos.[13] These inscriptions on these objects are presumably all non-administrative in nature and some may have been written prior to the Late Minoan IB destruction horizon. Alternatively, these objects may be evidence of the infrequent use of Linear A outside palace contexts for some time after the collapse of the Minoan palace administrations.

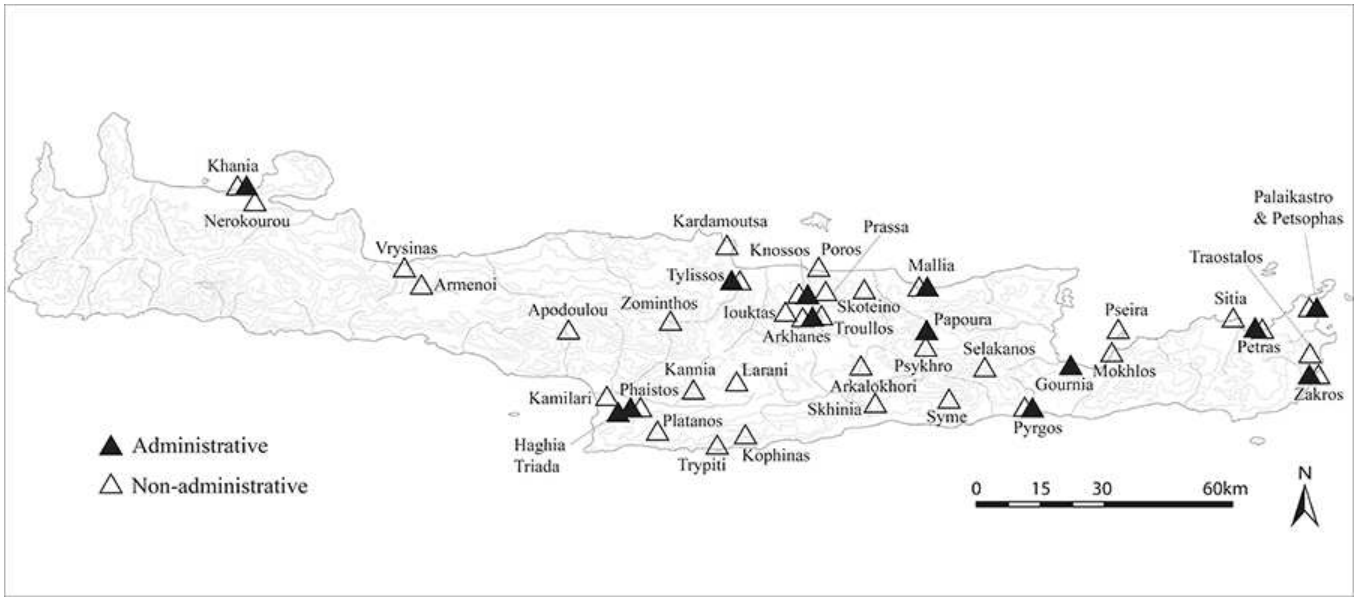

Figure 2. Geographical distribution of Linear A on Crete.

Source. Ester Salgarella, *Aegean Linear Script(s): Rethinking the Relationship Between Linear A and Linear B* (Cambridge, UK: Cambridge University Press, 2020), p. 2.

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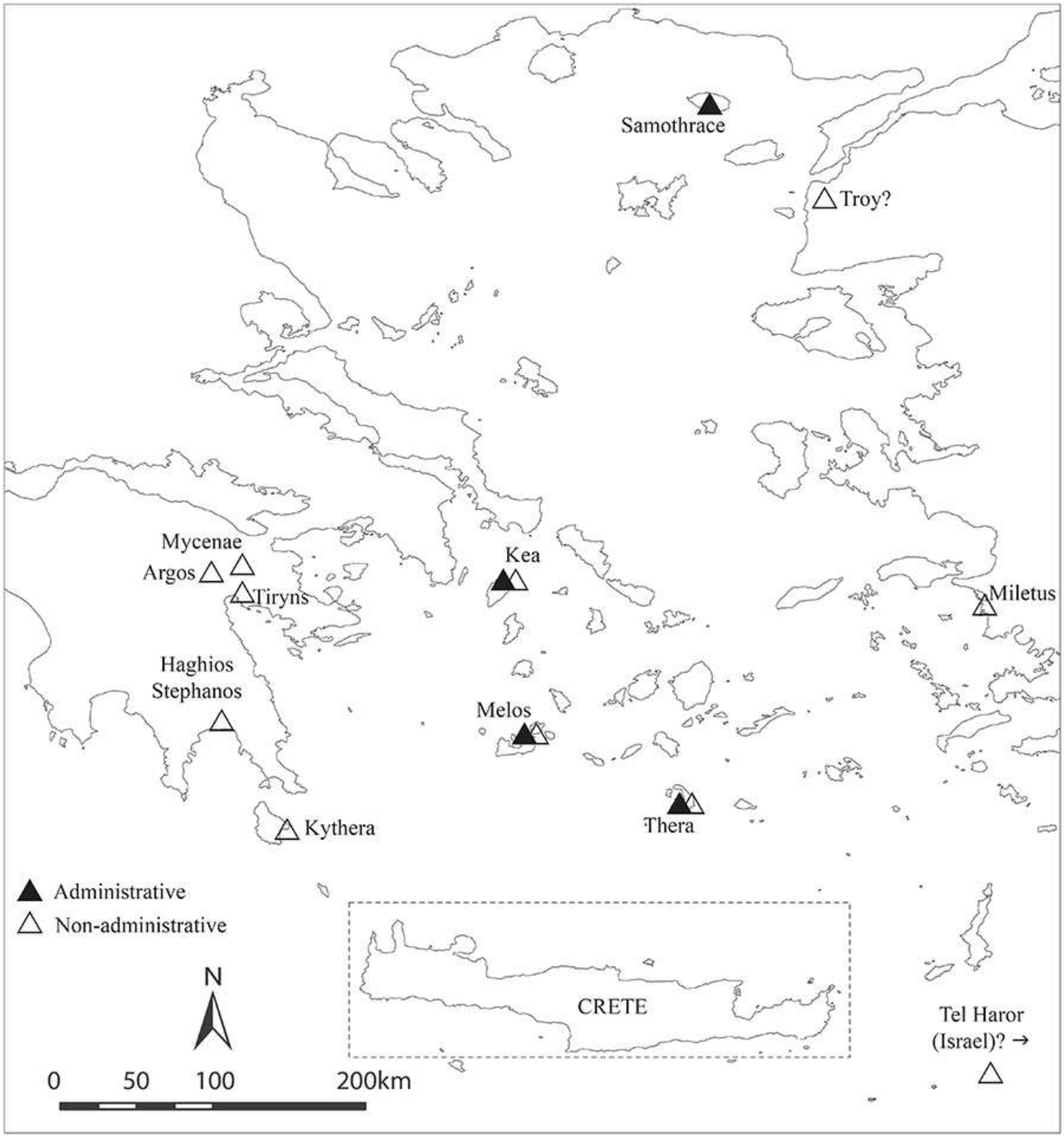

Figure 3. Geographical distribution of Linear A outside Crete.

Source. Ester Salgarella, *Aegean Linear Script(s): Rethinking the Relationship Between Linear A and Linear B* (Cambridge, UK: Cambridge University Press, 2020), p. 3.

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Document Types

Linear A is written on a variety of different materials (clay, stone, and metal) for either administrative purposes (records of economic transactions) or non-administrative purposes (mostly cultic; see Minoan and Mycenaean religion). To date, we have no evidence of monumental inscriptions, historiography, or diplomatic correspondence.

Administrative Contexts

Clay documents were the main media used for bookkeeping by palace administrations. Baked by accidental fires and thus available for us to read, these documents fall into the following categories:[14]

Tablets

Small page-shaped tablets were used to record the inflow and outflow of goods from palace centres and list mixed commodities, a single commodity, or a number of specialized commodities.[15] Tablets are often inscribed on both sides (turned over along the vertical axis) and many show traces of erasures, suggesting that the surface was erased and re-inscribed more than once. They were thus used for the temporary storage of information which before the tablets' reuse was transferred to perishable materials for longer-term preservation. Analysis of the textual information provided by tablets and other clay documents has permitted the tentative reconstruction of the workings of the Minoan economy and administrative machinery.[16]

Roundels

Small disks of clay inscribed on one or both sides and bearing seal impressions along their edge, roundels are understood to have functioned as records of transactions in palace centres. It has been suggested that the inscription (usually consisting of logograms) represents the commodity being dispatched from the storerooms and that the seal impressions indicate the quantity of the commodity in question.[17]

Sealings

Sealings were used to keep track of incoming and outgoing goods and prevent unauthorized manipulation of the sealed item. They are small lumps of clay, inscribed on one or more sides and with seal impressions, which were fastened to containers to secure their contents. There are three main types of sealed documents: nodules, noduli, and direct object sealings. Nodules come in different shapes: flat-based (pressed against folded parchment or papyrus), single-hole (attached to the end of a string, probably hanging from parchment or papyrus rolls), and two-hole (attached to commodities as labels, with a string passing through their length). Some flat-based nodules show imprints of fibre and string on their flat undersides, suggesting they were

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secured to folded papyrus and thus that Linear A was also written on perishable material, none of which has survived. Noduli (dome- or disc-shaped) are free-standing documents not attached to any support which are believed to have functioned as receipts—either for completed work or for commodities delivered to the palace centre—or as tokens exchanged for services or commodities.[18] Direct object sealings are pressed directly onto objects (jars and other containers, door pegs) to seal their contents. Although uninscribed, they show seal impressions, and some were found alongside Linear A inscribed tablets at Phaistos (Middle Minoan II).

Figure 4. Clay tablet from Zakros inscribed on both sides (ZA 10a *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=3&ce=540ecor2uudeumpibfksblm7ss9s2lsi&sp=192>*-b *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=3&ce=na45u6v17d4od89umoq8mt72o858q5r7&sp=194>*, 5.1 × 8.6 × 1.2 cm).

Source. Drawing by author.

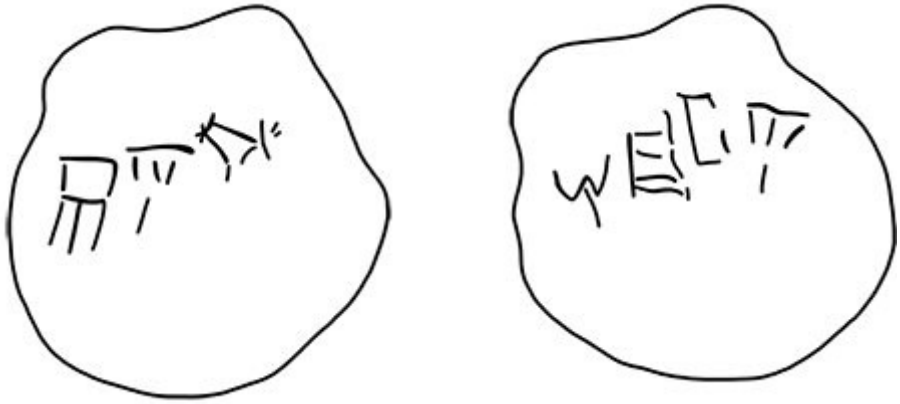

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Figure 5. Clay roundel from Haghia Triada inscribed on both sides (HT Wc 3008a-b <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=2&ce=na45u6v17d4od89umoq8mt72o858q5r7&sp=134>, 3.0 × 3.2 × 0.9 cm).

Source. Drawing by author.

Figure 6. Clay nodule from Khania inscribed on two sides (KH Wa 1001α-γ <https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=3&ce=na45u6v17d4od89umoq8mt72o858q5r7&sp=128>, 1.5 × 3.0 × 1.8 cm)

Source. Drawing by author.

Non-administrative Contexts

In non-administrative contexts we find a variety of inscribed objects whose purposes were primarily cultic but sometimes also functional.[19] The following media are attested:

1. Clay and stone vessels bearing short inscriptions (carved on stone, incised or painted on clay) found in both religious and domestic contexts (e.g., storage containers).
2. Votive stone libation tables and ladles bearing slightly longer inscriptions (variations of the so-called libation formula) and coming primarily from religious contexts (peak sanctuaries).
3. Metal objects (gold and silver pins, gold rings, bronze bowls, ceremonial double-axes) coming from burial contexts or cultic caves.
4. Various objects that do not fit neatly into any of the above categories: figurines, architectural blocks, loom-weights, seal-stones, graffiti on plaster.

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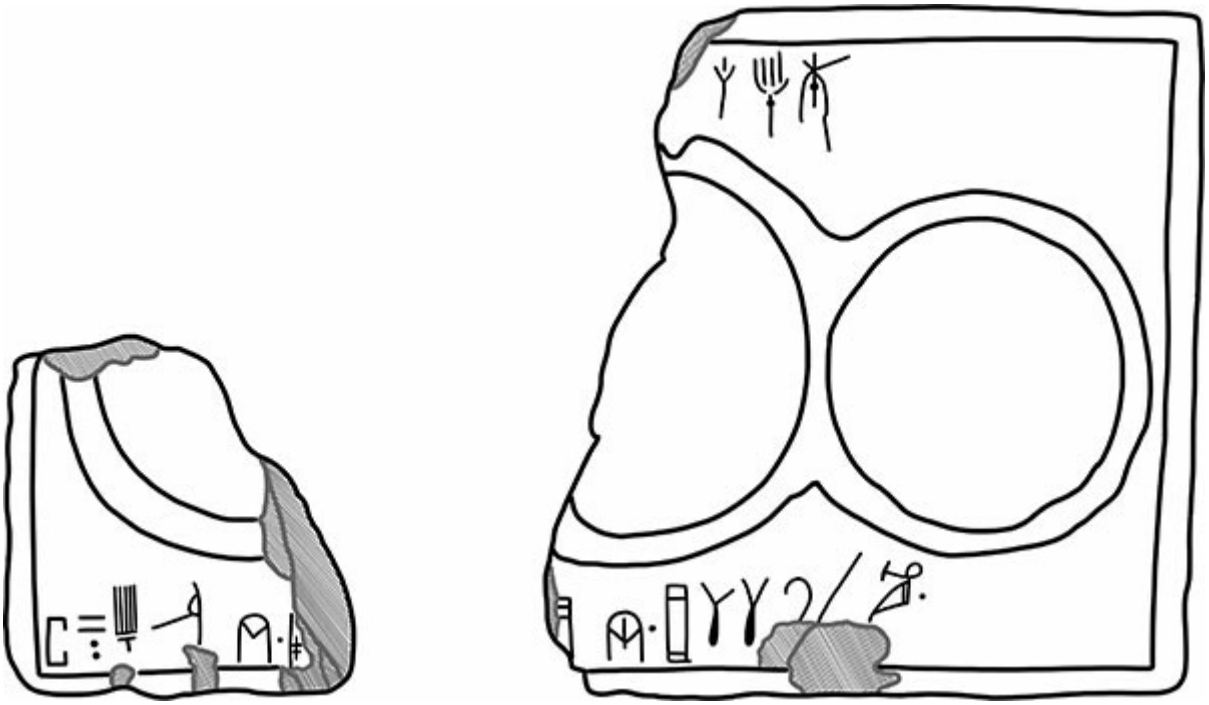

Figure 7. Libation table from Psykhro (PS Za 2 *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=4&ce=cooqm3a5q3s12vmvdqppfim4sk80ag09&sp=97>*).

Source. Drawing by author.

Administrative documents outnumber documents found in non-administrative contexts. On present evidence, clay sealings are the most common document type (nearly one thousand examples); next come clay tablets (some three hundred examples), followed by roundels (just over one hundred examples). There are around a hundred examples of incised clay and stone vessels (including libation tables and ladles). Inscriptions on other media such as metal and painted vessels are few and there is usually only a single example of each type.

The Minoan Language

At present, Minoan is believed to be an isolated language—that is to say, a language not affiliated to any known language family.[20] Based on the evidence of Linear A inscriptions, the Minoan language is likely to be agglutinative rather than inflected, since it appears to make extensive use of affixes (prefixes and suffixes), some of which have been successfully identified by scholars.[21] On the basis of etymology, scholars in recent decades have suggested that Minoan may be either a Semitic or Anatolian language (Palmer's Luwian theory is the most notable proposal in this respect) or may somehow be connected to Hurrian or Etruscan.[22] All attempts at identifying the language behind the script have so far proven untenable, with the evidence for each theory being insufficient. Current research has shifted toward statistical analyses and digital approaches to the Linear A corpus in an effort to identify the script's structural characteristics and any meaningful recurrent associations among sign-pairs and sign-sequences. Although attempts to find a linguistic affiliation for the Minoan language are currently at an impasse, it seems clear that: (a)

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Minoan may be a three-vowel system (/a/, /i/, /u/), since in Linear B most signs containing the vowel /o/ and one sign containing the vowel /e/ are new introductions rather than borrowings from Linear A;[23] (b) some of the Linear A syllabograms may represent consonantal clusters, as some evidence from Linear B suggests (e.g., the Linear B sign /nwa/, also found in both Linear A and Cretan Hieroglyphic, is not a Greek phoneme); (c) the substantial use of affixes suggests that these may have been used to express gender, case, or derivation, or may have had a prepositional meaning; and (d) the standard word order (based on evidence from the so-called libation formula) may have been verb-subject-object.[24] Despite such insights, Linear A still resists decipherment due to the relative paucity of the extant evidence, the short and formulaic nature of most inscriptions (which thus do not contain revealing syntactical structures), and the lack of an agreed-upon cognate language. Not a single bilingual inscription has yet been discovered.

Links to Digital Materials

Recueil des inscriptions en Linéaire A—Vol. 1 *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=1>*.

Recueil des inscriptions en Linéaire A—Vol. 2 *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=2>*.

Recueil des inscriptions en Linéaire A—Vol. 3 *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=3>*.

Recueil des inscriptions en Linéaire A—Vol. 4 *<https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=21&issue_number=4>*.

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SigLA—The Signs of Linear A: a palaeographical database *<https://sigla.phis.me/>*.

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Notes

1. Silvia Ferrara, Barbara Montecchi, and Miguel Valério, “The Relationship between Cretan Hieroglyphic and Linear A: A Palaeographic and Structural Approach,” *Biblioteca di Pasiphae* 16: 81–109); and Ester Salgarella, “Imagining Cretan Scripts: The Influence of Visual Motifs on the Creation of Script-Signs in Bronze Age Crete,” *Annual of the British School at Athens* 116: 63–94.

2. Louis Godart and Jean-Paul Olivier, *Recueil des inscriptions en linéaire A*, vol. 5 (Paris: Librairie Orientaliste Paul Geuthner, 1985), xxii–xxvii.

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3. This composite Linear A sign was also adopted in Linear B and used as a logogram for wool (**145/LANA*). It probably is the monogram of the Minoan word for wool, which later became the Greek μαλλός (*mallos*, "wool, fleece"). Hesychius explains that μάλλυκες (*mallykes*; s.v.) is the word for τρίχες (*trikhes*, "hair") on Crete. Hesiod uses the term μαλλός (*mallos*) for "fleece" (*Op*. 234).

4. Charts showing simple sign distribution are given in Ester Salgarella, *Aegean Linear Script(s): Rethinking the Relationship Between Linear A and Linear B* (Cambridge, UK: Cambridge University Press, 2020), 153–155 (figures 16 and 17).

5. Composite sign combinations and lists thereof are given in Salgarella, *Linear Script(s)*, 55–84; and Ester Salgarella, "Mix and Match: A Combinatory (Re-)classification of Linear A Signs," *TALANTA* (forthcoming).

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7. On the structural and palaeographic relationship between Linear A and Linear B and the script transmission process, see, most recently, Ester Salgarella, "Drawing Lines: The Palaeography of Linear A and Linear B," *Kadmos* 58 (2019): 61–92; and Salgarella, *Linear Script(s)*, where the author argues that Linear B not only emerged out of Linear A but actually is a graphic variant of the Linear A script, so that there is great continuity between the two.

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